Managing ICU Intubation Decisions for Optimal Patient Outcomes

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Question: When Can We Take the Breathing Tube Out?

- Too early $\implies$ risk of failure and reintubation $\implies$ possible pneumonia, death
- Too late $\implies$ complications from prolonged intubation (pneumonia, death); also pain/suffering of intubation

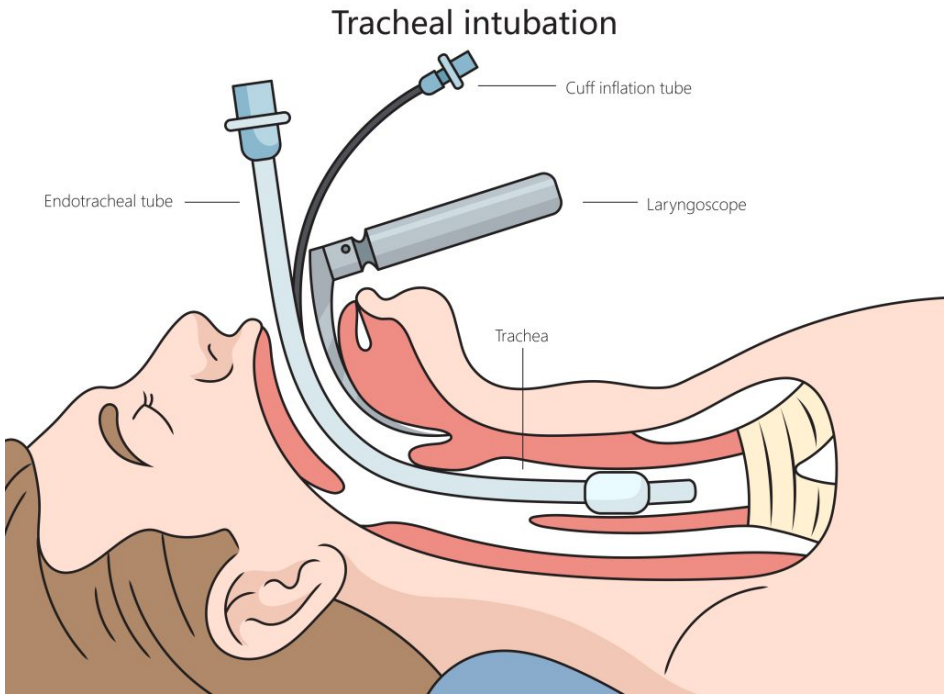

Figure 1: https://hayekmedical.com/2025/01/06/endotracheal-intubation-complications-and-risks/

The Extubation Decision

Assessment:

- Respiratory function:
  - oxygenation, respiratory rate, work of breathing
- Mental status:
  - alertness, ability to protect airway
- Cough strength, secretion clearance
- Underlying illness and clinical trajectory

objective data + clinical judgment + experience

Failed Extubation Rate

Failed Extubation = Extubated patient still requires mechanical assistance
$\implies$ Reintubation

- FE = failed extubation rate
- FE too high $\implies$ patient harm
- FE too low $\implies$ patient harm

Related Work

Krinsley JS, et al.: What is the optimal rate of failed extubation? Critical Care 2012, 16:111.

> *The mean (standard deviation) rates [of failed extubation] in interventional and observational studies are 15.7 (15.6) ($n = 55$) and 15.0 (14.0) ($n = 96$), respectively. . . . we choose the range of 5% to 10% as our proposed 'optimal' rate of failed extubation.*

- The wide range of observed failed extubation rates may indicate an opportunity for discovery.
- The authors' choice of 'optimal' rate is backed by professional judgement rather than data.

Current Work

Data are available:

- MIMIC IV database[1]
- eICU-CRD
- possible others

Can we find evidence to support an optimal rate for failed extubation?

[1] Access to these data requires project-specific approval.

Evaluation

Let y be some metric of patient health, such as 12-month survival.

- Goal:

$$\hat{y} = f(FE, x, \beta)$$

$$x = \text{observable patient variables (eg. } RSBI^2\text{)}$$

$$\beta = \text{other parameters (eg. physician index of caution)}$$

- Does FE in the 5-10% range optimize $\hat{y}$?
- Can we predict FE?

$$\widehat{FE} = g(x, \beta)$$

[2] Rapid Shallow Breathing Index: $RSBI = \frac{\text{resp. rate}}{\text{tidal vol.}}$

Toy Model/ Simulation

Before getting access to data, I developed a toy model in R and simulated data based on parameters suggested by Dr. Andrew Hersh, a pulmonary critical care physician.

This let me explore possile effects of various variables and parameters such as FE on y such as 12-month survival rates.

Preliminary investigations have yielded interesting theoretical results, but the models must be aligned with the data before any conclusions can be drawn.

Theoretical models and the real world

A mathematical model is a scientific hypothesis in precise language.

> *All models are wrong, but some are useful. -John Tukey, the topologist turned statistician who gave us box plots and taught us how to compute Fast Fourier Transforms*

To determine if our model is useful, we feed it data.

MIMIC-IV

Medical Information Mart for Intensive Care (MIMIC)-IV is a large deidentified dataset of patients admitted to the emergency department or an intensive care unit at the Beth Israel Deaconess Medical Center in Boston, MA.

- contains data for over 65,000 patients admitted to an ICU and over 200,000 patients admitted to the emergency department
- hosted at MIT
- cloud-scale - access is available via BigQuery (SQL)
- access requires project-specific credentialing via PhysioNet

For more information about this database, see the project website at `https://physionet.org/content/mimiciv/3.1/`.

Data Extraction

- Access to the MIMIC-IV data has been acquired. A data set specific to this question is being developed.
- $FE = 7.7\%$ in our data overall
- The data are cloud-scale and extracted from hospital electronic health records (EHR).
  - Errors and omissions in the EHR are common. Many of our desired variables were only available in a subset of cases.
  - Hence inference for missing or absent data is necessary.

Data Warehousing

A data set was built with 10,000+ observations of 20+ variables.

- Each observation is a patient who was extubated in the ICU.
- Variables include
  - patient age and gender;
  - principal and secondary diagnoses, and the diagnosis entered into the patients EHR closest to the time of intubation;
  - Surgical, Medical non-respiratory or Medical respiratory (inferred);
  - Charlson comorbidity score, SOFA[3];
  - norepinephrine equivalent dosage;
  - total ventilated hours (inferred in cases when intubation is not recorded);
  - the failed extubation rate of and number of extubations performed by the extubating caregiver; and
  - the date of death of the patient for those that did not survive 12 months past their extubation.

[3] Sequential Organ Failure Assessment `https://www.mdcalc.com/calc/691/sequential-organ-failure-assessment-sofa-score`

Data Summary

Table 1: Patient cohort summary (n = 11,801)

Metric	Value
Total patients	11,801
Age (years)	Min: 18, Median: 64, Max: 91
Sex	Female: 4,815 (40.8%), Male: 6,986 (59.2%)
Intubation Type	
Medical – Non-respiratory	2,984 (25.3%)
Medical – Respiratory	8,659 (73.4%)
Surgical	158 (1.3%)
Ventilation Duration (hrs)	Median: 68.8, Range: 1.1 to 998.7
SOFA Score	Median: 6, Range: 0 to 22
Charlson Index	Median: 5, Range: 0 to 19
Norepinephrine Dose (mcg/kg/min)	Median: 0.080, Range: 0 to 1.000
Caregiver FE rate	Median: 18.2%, Range: 0 to 94.7%
Extubations per caregiver	Median: 245, Range: 11 to 2,088
Failed extubation rate per patient	7.7%
Hospital mortality	26.7%
12-month mortality	44.4%

Data Mining

$\hat{y} = f(FE, x, \beta)$

- y is survival for 12 months post-extubation.
- FE is per caregiver.
- What are useful x and β?

Diagnoses certainly play an important role. May confound FE.

- Too many diagnoses to include directly.
- Group and select using a combination of medical knowledge[4] and machine learning[5].
  - Genitourinary, Neurologic and Endocrine $\implies$ **lower** caregiver FE.
  - Digestive and Infectious $\implies$ **higher** caregiver FE.

[4] ICD chapters and personal communication with physicians
[5] LASSO, Random Forest

Visualizing *FE* rates and survival

x-axis = caregiver *FE* y-axis = predicted survival (%) color = survival

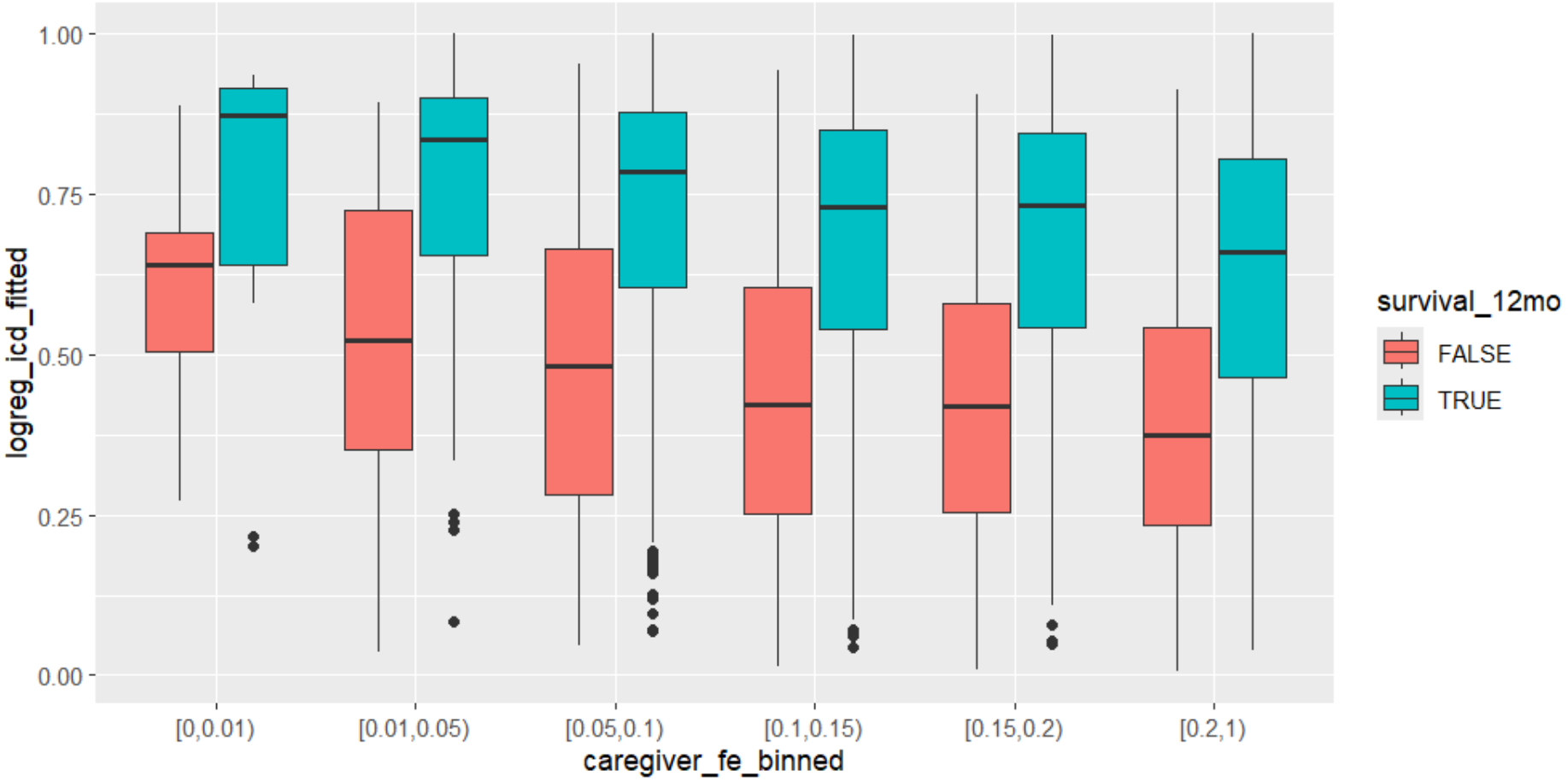

Figure 2: Caregiver Failed Extubation Rate and 12 Month Survival. Note the outliers who survived despite the odds.

Caregiver Experience

As `caregiver_n` $\to \infty$, $FE \to 10\%$.

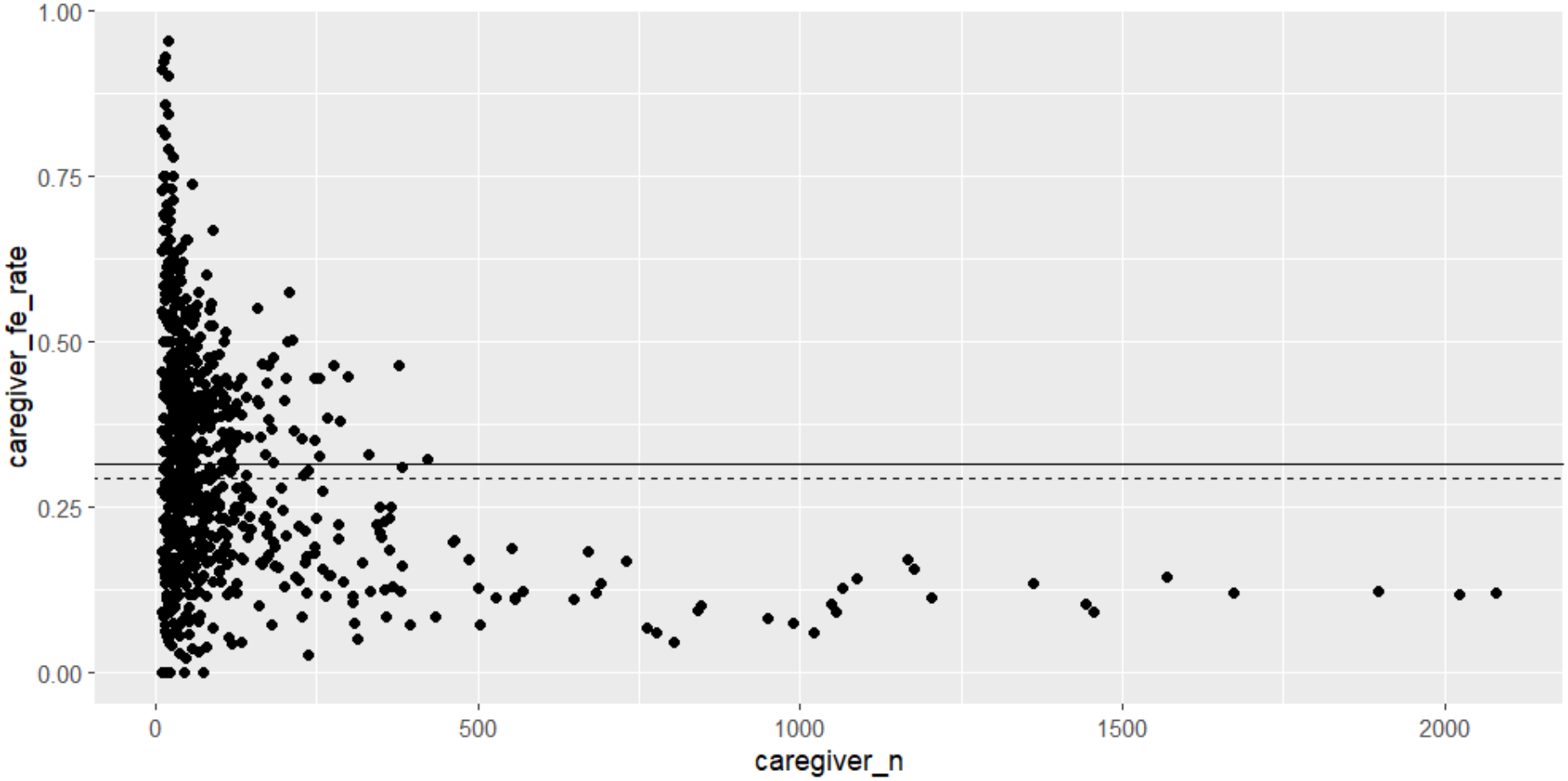

Figure 3: Caregiver Failed Extubation Rate and Number of Extubations

Logistic Regression Model for 12 month survival

Intubations were classified during data extraction as Surgical[6]/Medical Respiratory[7]/Medical Non-Respiratory.

- `glm(survival_12mo ~ poly(caregiver_fe_rate, 2) * (intubation_type + vent_hours + charlson_centered + norepinephrine_centered + sofa_centered + icd_digestive + icd_genitourinary + icd_respiratory + icd_neurologic + icd_special + icd_infectious + icd_circulatory + icd_pregnancy + icd_musculoskelatal), data = train_data, family = binomial)`

[6] surgical event charted within 12 hours of intubation

[7] respiratory diagnosis present in patient's chart

Model variables

- `poly(caregiver_fe_rate, 2)` $\rightarrow$ find FE maximizing survival?[8,9]
- Main effects for clinical and diagnostic variables.
- Indicator variables for the significant diagnosis groups.
  - The model that included the LASSO-suggested indicator variables had a ΔAIC of 116 compared to the best model without the diagnoses, which suggests that the diagnoses add useful information.
- Interactions between FE and all other variables.

[8] This would be the vertex of the parabola at $FE = -b/2a$, with a and b denoting the degree 2 and degree 1 coefficients on FE.

[9] A Gaussian or other more locally-supported function is probably more appropriate than a quadratic, but would complicate the present discussion.

Model Visualization (*FE*)

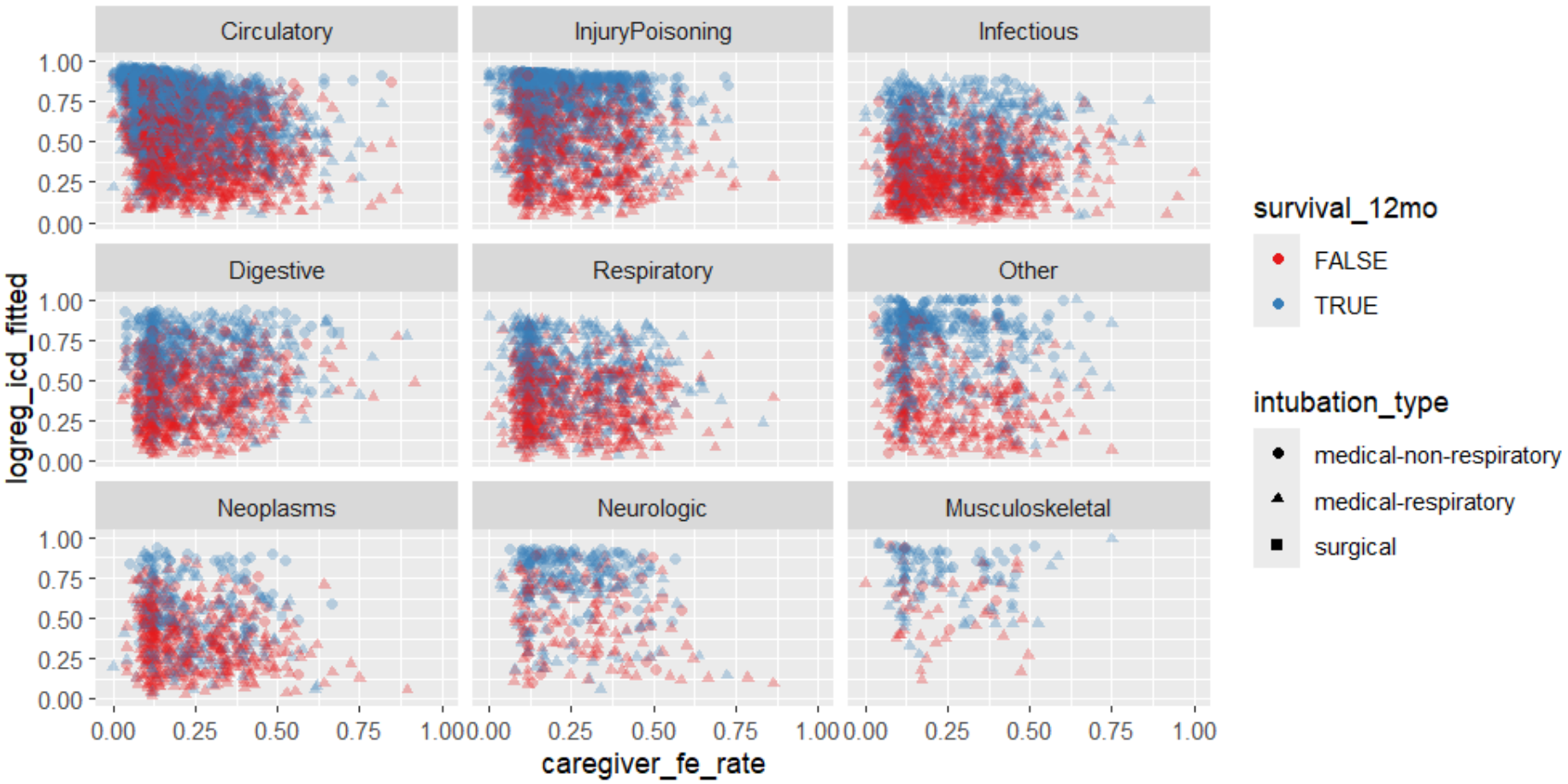

Figure 4: Predicted 12 month survival as a function of Diagnosis, Intubation Type, and Caregiver Failed Extubation Rate

Model Visualization (Charlson)

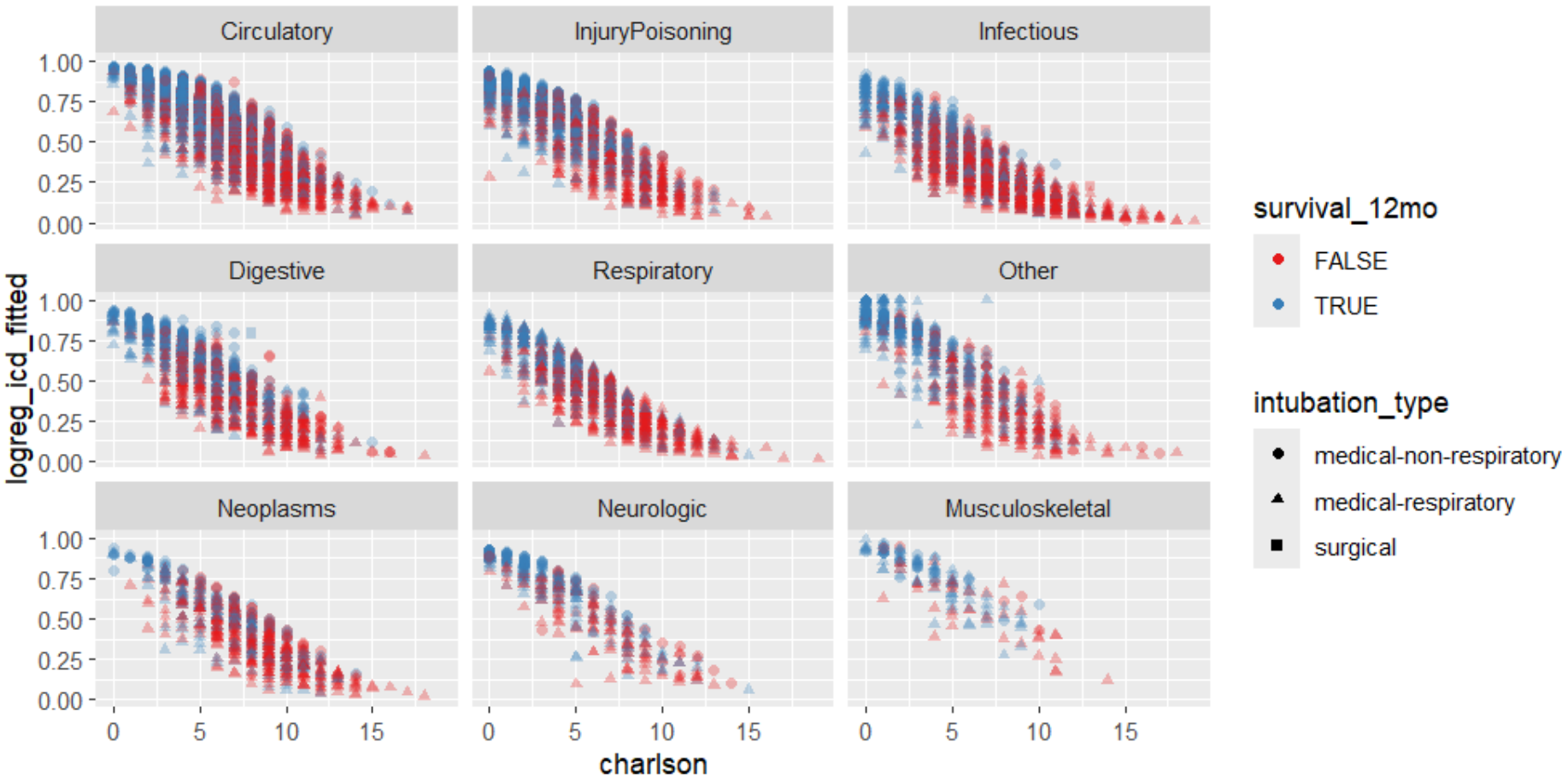

Figure 5: Predicted 12 month survival as a function of Diagnosis, Intubation Type, and Charlson Comorbidity Score

Model Visualization (SOFA)

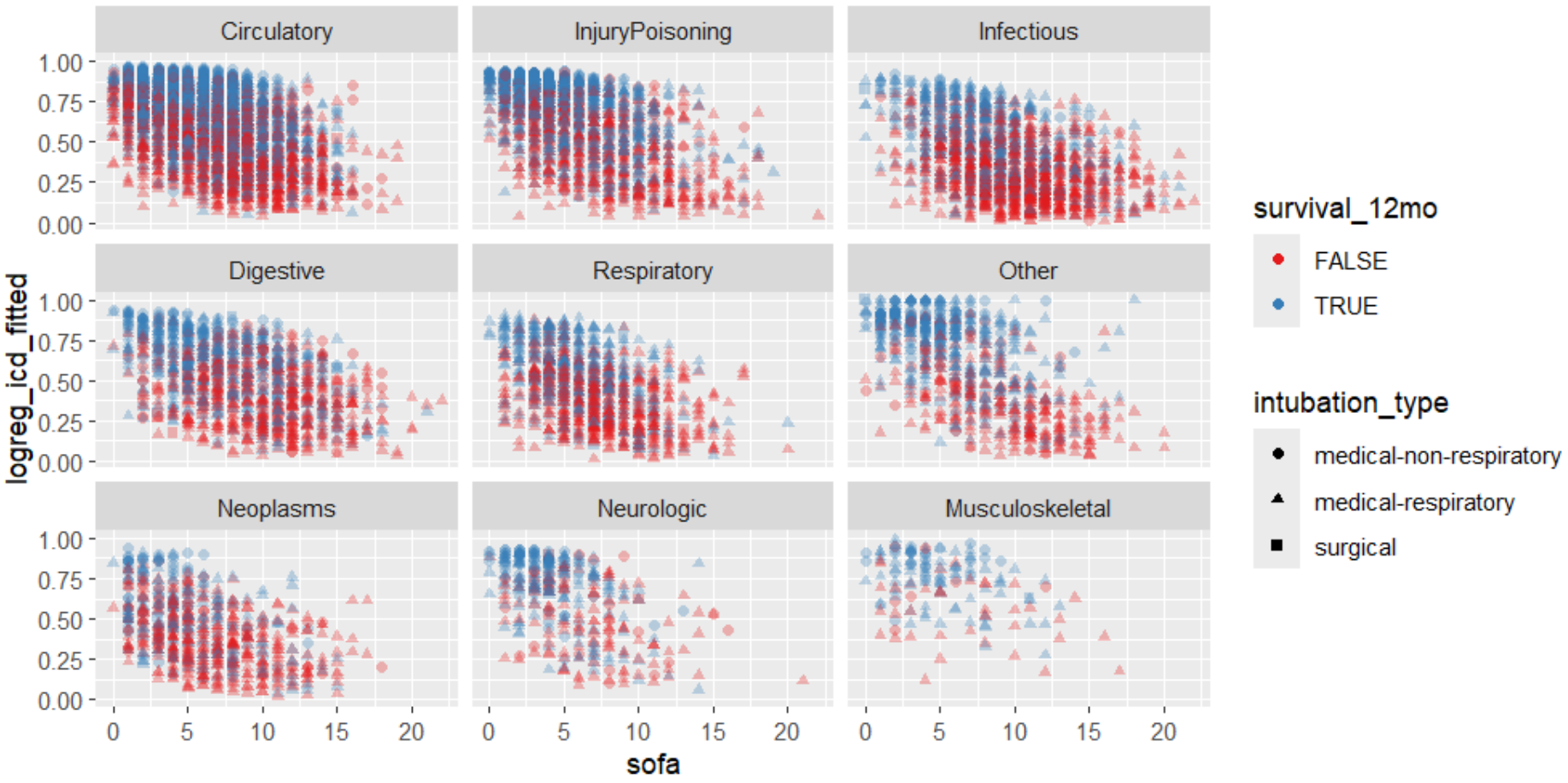

Figure 6: Predicted 12 month survival as a function of Diagnosis, Intubation Type, and SOFA

Model Evaluation

The model passes the visual sniff test.

To more objectively evaluate the form of the model, we performed 5-fold cross validation.

Table 2: 5-Fold Cross-Validation Performance Metrics

Metric	**Mean**	**SD**
Accuracy	0.707	0.012
Positive Predictive Value (Precision)	0.723	0.013
Negative Predictive Value	0.683	0.024
Sensitivity (Recall)	0.776	0.018
Specificity	0.619	0.023

Results

Observations:

- All clinical variables were found to be significant in the model, with the expected effects (e.g. longer vents are associated with lower survival).
- As a main effect, *FE* rate was found to be a significant predictor of 12 month survival...
- ...but the quadratic spline simply indicated a linear decrease in survival probability for increasing *FE* rather than an optimal *FE* that maximized estimated survival.
- See Figure 3. Caregivers with more ICU experience are more likely to have developed the necessary judgement required to achieve the goal of a 5-10% rate of *FE*.

More Results

- Diagnosis groups that the model found significant as main effects:
  - Endocrine - associated with better survival
  - Neurologic, External Causes - associated with worse survival
- Higher *FE* is associated with worse survival in Neurologic patients.

Lessons Learned: Caregiver Experience

Figure 3 was really interesting.

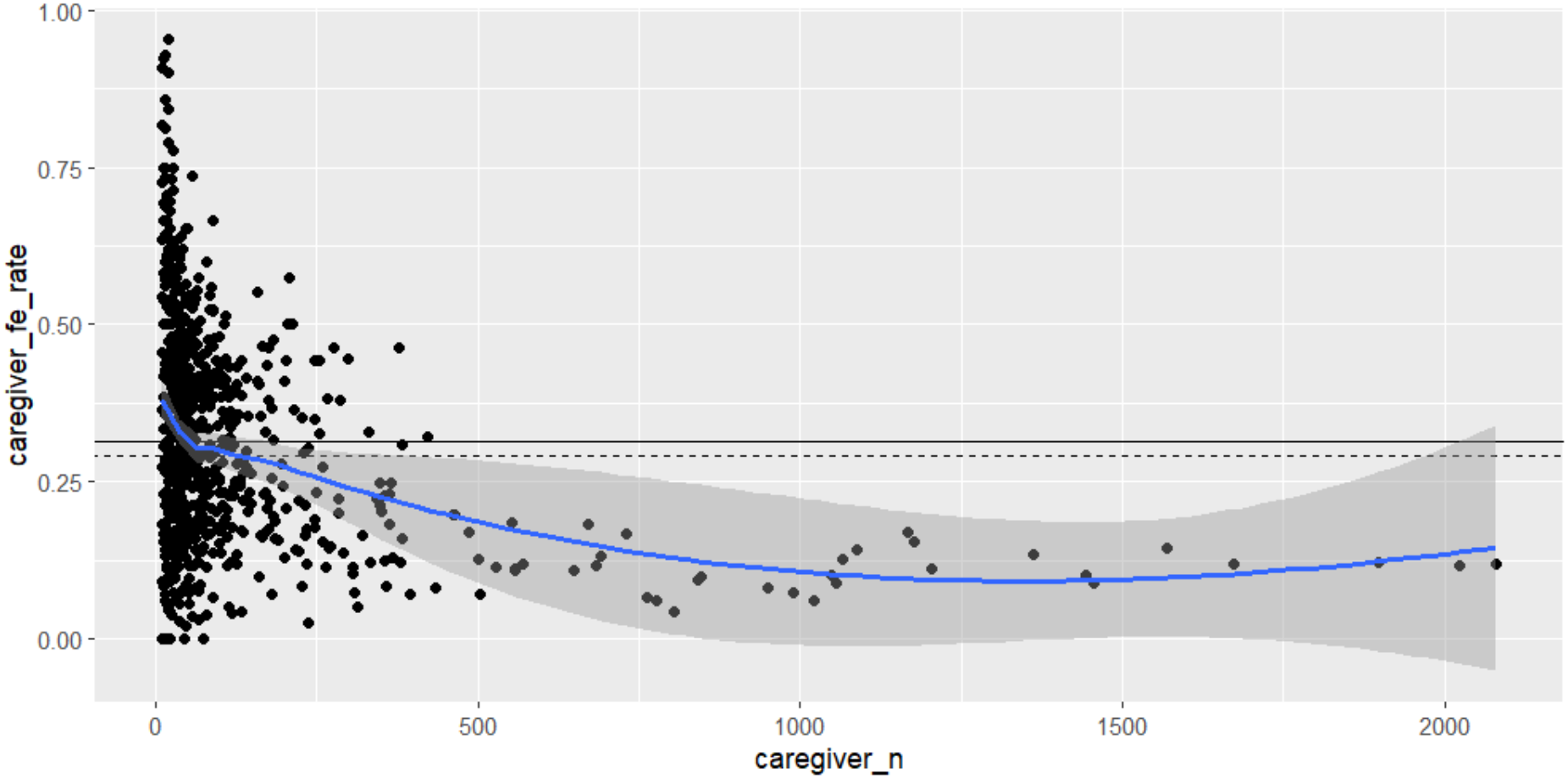

Figure 7: Caregiver Failed Extubation Rate and Number of Extubations

Caregiver Experience and Patient Survival

As `caregiver_n` $\to \infty$, 12-Month Survival remains typical.

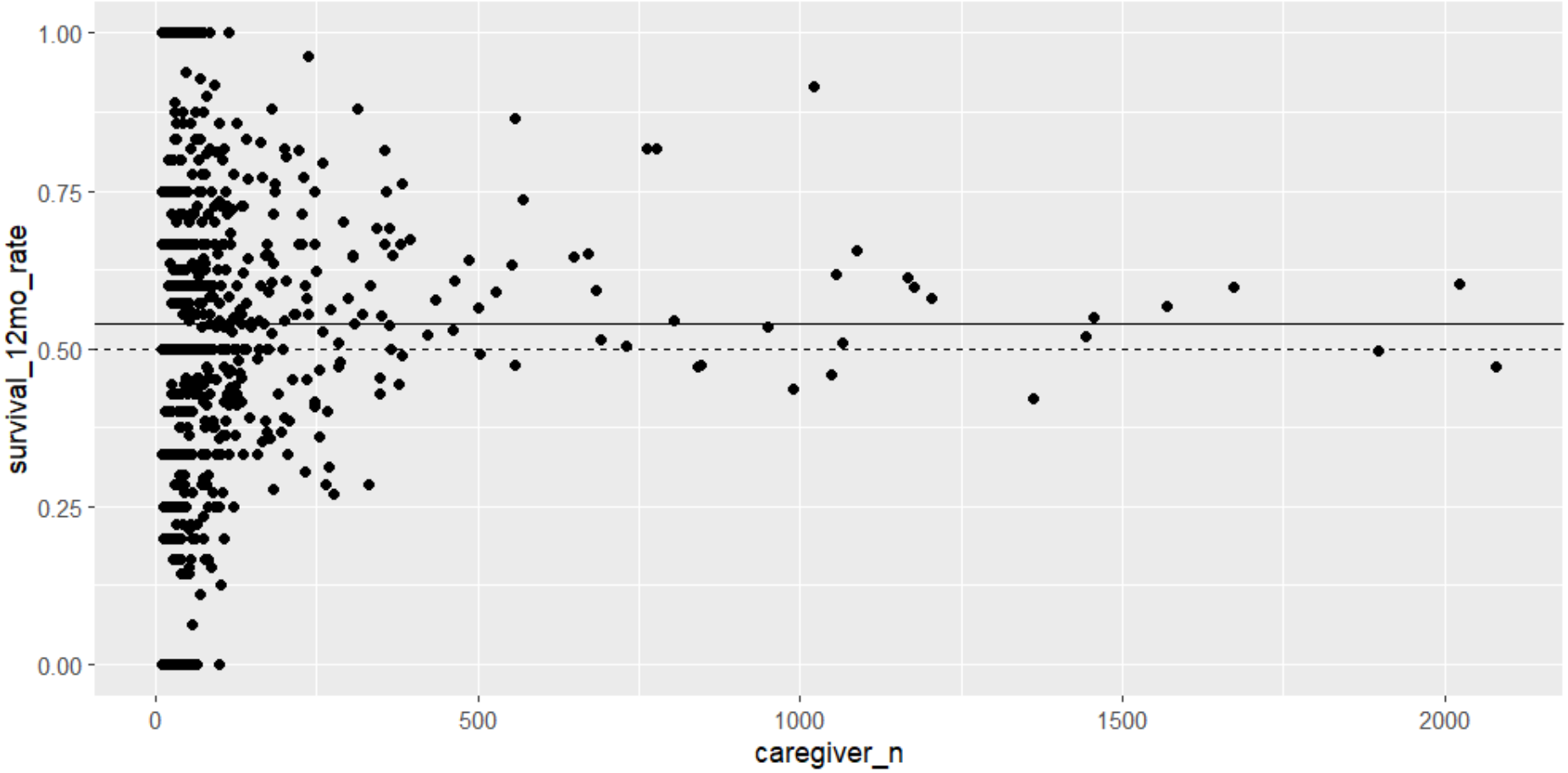

Figure 8: Caregiver 12-Month Survival Rate and Number of Extubations

Future Work

Next tasks:

- diagnoses $\rightarrow$ clustering and pattern analysis
- Better models or other metrics?
  - Mixed-effects models
  - In-hospital mortality
- Data $\rightarrow$ Simulation
- Contribute SQL that constructs our dataset to the MIMIC-Derived project.
- More ICUs. All of the MIMIC IV data are from one hospital in Boston.

References I

- Ary L. Goldberger, Luis A. N. Amaral, Leon Glass, Jeffrey M. Hausdorff, Plamen Ch. Ivanov, Roger G. Mark, Joseph E. Mietus, George B. Moody, Chung-Kang Peng, and H. Eugene Stanley, *Physiobank, physiotoolkit, and physionet: Components of a new research resource for complex physiologic signals*, Circulation **101** (2000), no. 23, e215–e220, Online.
- Alistair Johnson, Laura Bulgarelli, Tom Pollard, Benjamin Gow, Benjamin Moody, Steven Horng, Leo Anthony Celi, and Roger Mark, *Mimic-iv (version 3.1)*, `https://doi.org/10.13026/kpb9-mt58`, 2024, PhysioNet.
- Alistair E. W. Johnson, Laura Bulgarelli, Lu Shen, and et al., *Mimic-iv, a freely accessible electronic health record dataset*, Scientific Data **10** (2023), 1.

References II

- James Krinsley, Praveen Reddy, and Abid Iqbal, *What is the optimal rate of failed extubation?*, Critical care (London, England) **16** (2012), 111.
- Andrew Hersh MD, personal communication.

The End

Thanks!

Questions?